

SPACE SHOOTER GAME

Description

A 2D space shooter where a player-controlled spaceship dodges and shoots falling enemy ships. The player-controlled ship and alien ships are an

Game Goal:

Try to kill as many aliens as possible to keep the world in balance

Game End:

The game ends when the player collides with an alien spaceship.

Game Challenges:

More aliens spawn each time player's kill count crosses a set threshold. Aliens start shooting back at player after reaching certain level is reached.

Design

Game States

- game_is_running (True): Controls main game loop
- player_has_health (True): Shows whether player is still alive

Classes (Objects Used In Game)

- Player : Spaceship the gamer gets to control
 - *follow_mouse_pointer()* : Lets player object follow the mouse pointer across visible game screen
 - *shoot_EB()* : Create and fire an energy blast
 - *Shoot_Missile()* : Create and throw a *missile* projectile at enemy
- Alien (Enemy) : Enemy mob the player must destroy
 - *Move()* : Randomly move around defined area on screen
 - *Shoot()* : Create and throw a projectile at the player
 - *Disappear()* : Remove enemy from screen
- AlienManagers : Manages the enemy mobs the player must destroy
- Projectile (Missile & Energy Blast) : The bullets the player and enemies can fire
 - *Move()* : Move straight ahead in the direction it was facing
 - *Explode()* : Display explosion animation
 - *Disappear()* : Remove the object from the screen
- BulletManager : Manages all the bullets (EB / Missile) created by the player
 - *add_bullet()*: Creates an EnergyBlast object and stores it in a list
 - *update_fired_bullets()*: Moves all fired EnergyBlast objects across the screen
 - *_remove_hidden_bullets()*: Remove any EB objects that are off-screen

- Scoreboard : Keeps track of number of aliens killed by player
 - Update_Score() : *Update the user's kill count in local variable*
 - Record_Score() : *Write user's kill count to file or store in database (later on)*
- Environment (Background)
 - Animate() : *Loop through the moving background animation*

Assets

- Background animation frames
- Player spaceship animation frames
- Alien Ship1 animation frames
- Alien Ship2 animation frames
- Energy Blast animation frames
- Missile animation frames
- Explosion Fx animation frames
- Player spaceship flying sound (<https://pixabay.com/sound-effects/search/spaceship%20ambience/>)
 - Sound Effect by Paweł Sychała from Pixabay
- Alien1 and Alien2 sounds (later on)
- Energy blast sound Fx (<https://pixabay.com/sound-effects/search/laser%20gun/>)
 - Sound Effect by freesound_community from Pixabay
- Missile launch sound Fx (<https://pixabay.com/sound-effects/search/tank%20fire/>)
- Explosion sound Fx
- Background Music (<https://pixabay.com/music/search/sci%20fi/> and)
 - Music by mariokhol from Pixabay
 - Space Ambience: Youtube Guy (Remember to cite him too!)

Roadmap

- Create player object and let it follow the mouse pointer across the screen
 - Convert that to a class and write necessary code for basic movements
- Create energy blast object and let it move up the screen until it disappears
 - Convert that to a class and write necessary code for animation and disappearance.
- Create space animation background to loop while game is running (Thank you GPT 🤖)
- Add necessary music and sound effects for current progress
 - Energy blast released sFx
 - Player spaceship flying sFx
 - Space background music
- Create test enemy object with animations and let it move down the screen. move randomly about the screen for a few seconds, and then fall down towards the user.
 - Alien should stop after a certain y-coord. Then randomly move along the x-axis (or y) for a few seconds.
 - Alien should move down to the ground.
 - Convert that to a class and write necessary code for implemented movements.

- Implement alien logic into main game logic
 - Create test energy blast and alien collision logic
 - Create test player object and alien collision logic
 - Create test explosion animation effect
 - Implement test explosion effect when collision is detected
 - Convert that to a class and write necessary code for implementation
- Create Scoreboard to track number of aliens player kills

Links to Assets

- Energy blast sound Fx (<https://pixabay.com/sound-effects/search/laser%20gun/>)
 - Sound Effect by [freesound_community](https://pixabay.com/users/freesound_community-46691455/?utm_source=link-attribution&utm_medium=referral&utm_campaign=music&utm_content=105781) from [Pixabay](https://pixabay.com/sound-effects/?utm_source=link-attribution&utm_medium=referral&utm_campaign=music&utm_content=105781)
- Background Music (<https://pixabay.com/music/search/sci%20fi/>)
 - Music by [mariokhol](https://pixabay.com/users/mariokhol-17683309/?utm_source=link-attribution&utm_medium=referral&utm_campaign=music&utm_content=268026) from [Pixabay](https://pixabay.com/music/?utm_source=link-attribution&utm_medium=referral&utm_campaign=music&utm_content=268026)
- Player spaceship flying sound (<https://pixabay.com/sound-effects/search/spaceship%20ambience/>)
 - Sound Effect by [Paweł Spychała](https://pixabay.com/users/audiopapkin-14728698/?utm_source=link-attribution&utm_medium=referral&utm_campaign=music&utm_content=296518) from [Pixabay](https://pixabay.com/sound-effects/?utm_source=link-attribution&utm_medium=referral&utm_campaign=music&utm_content=296518)